

The Post-*Shelby* VRA Landscape: Section 5’s Demise, Section 2’s Evolution, and Algorithmic Redistricting as Transparency Substitute

Giovanni Della-Libera

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Abstract

Shelby County v. Holder (2013) struck down the coverage formula of Section 4(b) of the Voting Rights Act, rendering Section 5 preclearance a dead letter without formally repealing it. The nine states and numerous counties that had been required to obtain federal approval before changing any voting procedure were freed from that obligation overnight. Congressional attempts to update the coverage formula—the John Lewis Voting Rights Advancement Act and its predecessors—have not succeeded as of 2026. The post-*Shelby* VRA enforcement landscape is therefore one in which the most powerful ex ante minority voting rights tool has been eliminated, leaving Section 2 vote-dilution litigation as the primary federal remedy for discriminatory redistricting.

This paper maps the post-*Shelby* landscape for redistricting practitioners. We analyze the constitutional basis and consequence of *Shelby County*, trace the evolution of Section 2 from an auxiliary backstop to the primary federal enforcement mechanism, and assess the doctrinal implications of three subsequent Supreme Court decisions: *Abbott v. Perez* (2018), *Brnovich v. Democratic National Committee* (2021), and *Allen v. Milligan* (2023). We then argue that algorithmic redistricting—specifically the BISECT pipeline with its full SHA audit chain—provides a transparency substitute for the Section 5 preclearance process: a reproducible, neutral reference plan against which enacted maps can be measured ex post, serving the same evidentiary function for Section 2 plaintiffs that Section 5 non-retrogression review served ex ante for covered jurisdictions. The BISECT VRA mode (`--weights-override vra-aligned`) is designed to satisfy Section 2’s *Gingles* requirements without the preclearance submission process that Section 5 required, making it fully adapted to the post-*Shelby* legal environment.

Keywords

Shelby County v. Holder, Section 5 Preclearance, Section 2 Vote Dilution, *Thornburg v. Gingles*, *Allen v. Milligan*, *Brnovich v. DNC*, Algorithmic Redistricting, BISECT, VRA Enforcement, Transparency

1 Introduction: The Post-*Shelby* Enforcement Gap

1.1 June 25, 2013

On June 25, 2013, the Supreme Court decided *Shelby County v. Holder*.¹ By a 5–4 vote, the Court invalidated the coverage formula in Section 4(b) of the Voting Rights Act of 1965—the provision that designated which states and counties were subject to Section 5 preclearance—on the ground that it was based on data from 1964–1972 and no longer reflected current conditions sufficient to justify its departure from the equal sovereignty of the states.

Section 5 itself was not struck. The majority opinion by Chief Justice Roberts acknowledged that Congress could enact a new coverage formula based on current data, and it left Section 5’s operative requirements in place. But without a valid coverage formula, no jurisdiction is covered. Section 5 became a dead letter within hours of the decision.

The practical consequences were swift. Texas, which had been one of the nine fully covered states, announced the same day that it would implement a strict voter ID law that had previously been rejected under preclearance. North Carolina enacted an omnibus voting restrictions bill within months. Other former covered states moved to implement redistricting plans that had been blocked or conditioned under the preclearance process.

1.2 What the Post-*Shelby* Landscape Looks Like

Thirteen years after *Shelby County*, the VRA enforcement landscape for redistricting has stabilized around Section 2. The *Thornburg v. Gingles*² three-part test—geographic concentration, political cohesion, and majority bloc voting—remains the operative standard for redistricting challenges. Section 2 cases proceed ex post, after maps are enacted, through federal court litigation that can take years. The structural asymmetry between Section 5 (ex ante, administrative, faster) and Section 2 (ex post, judicial, slower) is the defining feature of the post-*Shelby* enforcement environment.

¹*Shelby County v. Holder*, 570 U.S. 529 (2013).

²*Thornburg v. Gingles*, 478 U.S. 30 (1986).

1.3 Algorithmic Redistricting in the Post-*Shelby* World

This paper argues that algorithmic redistricting fills part of the gap that Section 5’s demise created. Section 5 required covered jurisdictions to prove, before implementation, that a proposed change did not retrogress minority voting strength. This ex ante proof requirement created a paper trail and a neutral benchmark: the existing plan provided the non-retrogression baseline. Without Section 5, no such proof is required before enactment, and no neutral baseline exists unless one is constructed.

The BISECT pipeline produces a neutral baseline by construction. Because it operates without partisan data—it uses Census demographics, geographic adjacency, and population counts, but not election results or voter registration data—the BISECT plan reflects what a principled, non-partisan redistricting process would produce. The gap between a BISECT plan’s BVAP distribution and an enacted plan’s BVAP distribution measures the portion of minority vote dilution that is attributable to mapmaker choice rather than geography. This is precisely the comparison that Section 2 litigation requires—and precisely the comparison that Section 5 would have caught ex ante.

1.4 Overview

Section ?? describes the pre-*Shelby* Section 5 framework and the *Shelby County* holding. Section ?? traces Section 2’s evolution as the primary post-*Shelby* enforcement mechanism, including the effects of *Abbott v. Perez* (2018) and *Brnovich v. Democratic National Committee* (2021). Section ?? analyzes *Allen v. Milligan* (2023) and state VRA analogs. Section ?? develops the algorithmic redistricting as transparency substitute argument. Section ?? concludes.

2 Section 5 Before and After *Shelby County*

2.1 The Pre-*Shelby* Section 5 Framework

Section 5 of the Voting Rights Act required “covered” jurisdictions—those designated under Section 4(b)’s coverage formula—to obtain federal approval (“preclearance”) before implementing any change to a voting law or procedure. The coverage formula in Section 4(b) was based on two criteria: (1) use of a “test or device” (such as a literacy test) as a condition of voting as of November 1964, 1968, or 1972, and (2) voter registration or turnout below 50% in the presidential elections of those years.

The nine states fully covered under Section 5’s original formula were Alabama, Alaska,

Arizona, Georgia, Louisiana, Mississippi, South Carolina, Texas, and Virginia. Parts of additional states were also covered, including counties in California, Florida, Michigan, New Hampshire, New York, North Carolina, and South Dakota.

Preclearance required covered jurisdictions to demonstrate to the Department of Justice (administrative preclearance) or the U.S. District Court for the District of Columbia (judicial preclearance) that any proposed change in voting procedure had neither the purpose nor the effect of retrogressing minority voting strength. The “non-retrogression” standard, articulated in *Beer v. United States*,³ meant that covered states could change their voting procedures and redistrict, but they could not reduce the minority electoral opportunity they had previously afforded.

For redistricting, Section 5 operated as an ex ante floor: each redistricting cycle, covered states had to demonstrate that the new plan did not reduce the number of majority-minority districts or otherwise retrogress minority voting strength. This generated extensive DOJ review of redistricting plans, with conditions and objections imposed when plans fell below the non-retrogression standard.

2.2 *Shelby County v. Holder* (2013): The Holding

Chief Justice Roberts, writing for the five-Justice majority, held that Section 4(b)’s coverage formula was unconstitutional under the Fifteenth Amendment’s enforcement clause and the equal sovereignty principle of the states.⁴ The majority’s reasoning rested on two premises.

First, the coverage formula was based on data from 1964–1972 and had not been updated in the 2006 reauthorization, which simply extended the existing formula for twenty-five years. The majority held that this was constitutionally impermissible: the formula must be “sufficiently related to the problem it targets” and the current problem of minority vote suppression in formerly covered states was, in the majority’s view, not demonstrably worse than in non-covered states using 2012 data.⁵

Second, the majority applied the equal sovereignty principle—the doctrine that states must be treated equally by federal legislation—and concluded that Section 5’s differential treatment of covered states could not be sustained without a coverage formula calibrated to current conditions.⁶

The dissent by Justice Ginsburg (joined by Justices Breyer, Sotomayor, and Kagan) argued that the majority had evaluated the VRA against its own success: the coverage

³*Beer v. United States*, 425 U.S. 130 (1976).

⁴*Shelby County v. Holder*, 570 U.S. 529, 544 (2013) (“The Fifteenth Amendment is not designed to punish for the past; its purpose is to ensure a better future.”).

⁵*Id.* at 556.

⁶*Id.* at 554–56.

formula’s jurisdictions had fewer Section 2 violations not because discrimination was absent but because Section 5 was deterring it.⁷

2.3 Section 5 After *Shelby County*: A Dead Letter

Section 5 remains in the U.S. Code at 52 U.S.C. § 10304. But without a valid coverage formula in Section 4(b), no jurisdiction can be “covered” under Section 5. The Supreme Court left Section 5 intact as statutory text while making it unenforceable, in the expectation that Congress would enact a new coverage formula.

Congress has not done so. The John Lewis Voting Rights Advancement Act, which would update the coverage formula based on a new methodology tracking voting rights violations in the preceding twenty-five years, has passed the House of Representatives but has not cleared the Senate as of 2026. The most powerful provision of the VRA—the one that operated *ex ante*, with an administrative process that was faster and cheaper than federal litigation—thus remains unenforceable.

2.4 Immediate Post-*Shelby* Redistricting Consequences

The immediate redistricting consequence of *Shelby County* was the implementation of plans that had been held under preclearance review or that would have been blocked under preclearance. Texas implemented an interim congressional plan drawn by a three-judge court during Section 5 litigation; after *Shelby County*, the state enacted its own plan. North Carolina redrew its congressional map in ways that federal courts later found in *Covington v. North Carolina* to involve racial predominance in violation of the Equal Protection Clause.⁸

These consequences illustrate the prophylactic function that Section 5 had served: it did not merely remedy violations after the fact but deterred them before enactment. Once that deterrent was removed, some former covered states moved quickly to implement maps that would face years of Section 2 litigation.

⁷*Id.* at 591–92 (Ginsburg, J., dissenting).

⁸*Covington v. North Carolina*, 316 F.R.D. 117 (M.D.N.C. 2016) (finding that 28 of 170 state legislative districts were racially gerrymandered); see also *North Carolina v. Covington*, 581 U.S. 1 (2017) (per curiam).

3 Section 2’s Evolution as the Primary Enforcement Mechanism

3.1 Section 2 as the Primary Post-*Shelby* Tool

Section 2 of the Voting Rights Act, 52 U.S.C. § 10301, prohibits any voting qualification, standard, practice, or procedure that “results in” the denial or abridgment of the right to vote on account of race or color. As amended in 1982, Section 2 employs a “results” standard: plaintiffs need not prove discriminatory intent but only that the challenged practice has a discriminatory effect. In the redistricting context, Section 2 is primarily enforced through the vote-dilution framework of *Thornburg v. Gingles*.⁹

Gingles established three preconditions for a Section 2 claim that a redistricting plan unlawfully dilutes minority voting strength by failing to create a majority-minority district:

1. *Numerosity and compactness*: The minority group is “sufficiently large and geographically compact to constitute a majority in a single-member district.”
2. *Political cohesion*: The minority group is “politically cohesive”—members of the group tend to vote together.
3. *Majority bloc voting*: The white majority votes sufficiently as a bloc to defeat the minority group’s preferred candidates in the absence of a majority-minority district.

If all three preconditions are met, the court examines the “totality of circumstances” to determine whether the challenged plan “results in” minority vote dilution. The analysis includes, but is not limited to, the history of discrimination, the extent of racially polarized voting, the use of unusually large election districts, and the lack of success of minority candidates in the jurisdiction.

3.2 *Abbott v. Perez* (2018) and the Intent Bar

*Abbott v. Perez*¹⁰ addressed the Texas redistricting litigation that had been running since the 2012 cycle. The Supreme Court, in an opinion by Justice Alito, reversed the district court’s findings that the majority of challenged Texas congressional and state legislative districts violated the VRA or the Equal Protection Clause. The Court imposed a high burden on plaintiffs seeking to prove discriminatory intent, requiring evidence sufficient to overcome the presumption of legislative good faith.

⁹*Thornburg v. Gingles*, 478 U.S. 30 (1986).

¹⁰*Abbott v. Perez*, 585 U.S. 579 (2018).

For Section 2 redistricting claims based on discriminatory intent (as opposed to the results-based dilution analysis of *Gingles*), *Abbott v. Perez* makes it substantially harder to prevail. But the decision did not alter the *Gingles* results-based framework; it addressed the intent prong that applies when plaintiffs allege racial gerrymandering under the Equal Protection Clause rather than vote dilution under Section 2.

3.3 *Brnovich v. Democratic National Committee* (2021)

*Brnovich v. Democratic National Committee*¹¹ is the most significant post-*Shelby* weakening of Section 2, but its scope is limited to vote-denial claims. The case involved two Arizona election rules: a policy of not counting out-of-precinct provisional ballots and a law prohibiting ballot collection by third parties. The five-Justice majority, per Justice Alito, held that these rules did not violate Section 2’s results test, and in doing so articulated a multi-factor framework for evaluating non-redistricting Section 2 claims.

The *Brnovich* framework applies to Section 2 vote-denial claims (challenges to election rules that burden minority voters’ ability to vote) rather than Section 2 vote-dilution claims (challenges to redistricting plans that dilute minority voting strength). Justice Kagan’s dissent, joined by Justices Breyer and Sotomayor, argued that the majority had effectively created a new test for vote-denial claims that was less demanding of defendants than the statutory text required.

For redistricting practitioners, *Brnovich*’s key significance is what it *did not* do: it expressly preserved the Section 2 vote-dilution framework for redistricting claims, and it did not alter the *Gingles* analysis. The Court’s opinion noted that the two categories of Section 2 claim—vote denial and vote dilution—are analyzed differently, and that the *Brnovich* analysis was directed at vote-denial claims only.¹² The *Gingles* framework thus survived *Brnovich* intact for redistricting purposes.

3.4 Congressional Inaction: The Coverage Gap Persists

The John Lewis Voting Rights Advancement Act (H.R. 4), named for the late Georgia congressman and civil rights icon, would establish a new coverage formula for Section 5 based on a different methodology: jurisdictions with more than a threshold number of voting rights violations in the preceding twenty-five years would be covered. The Act passed the House in December 2019 and again in August 2021 but has not received a Senate vote as of 2026.

¹¹*Brnovich v. Democratic National Committee*, 594 U.S. 647 (2021).

¹²*Brnovich*, 594 U.S. at 680 n.19.

The failure to restore Section 5 preclearance means that the 2020 and 2030 redistricting cycles in former covered states proceed entirely under Section 2, without the ex ante review that Section 5 provided. Former covered states retain no federal obligation to submit redistricting plans for review; minority plaintiffs must file suit after enactment and litigate for years before courts may order remedial maps.

This asymmetry—ex ante administrative review under Section 5 versus ex post judicial review under Section 2—is the defining structural feature of the post-*Shelby* redistricting environment for the foreseeable future.

4 *Allen v. Milligan* (2023) and State VRA Analogs

4.1 *Allen v. Milligan* (2023): *Gingles* Reaffirmed

*Allen v. Milligan*¹³ is the most important post-*Shelby* Supreme Court decision on redistricting and VRA Section 2. The case arose from Alabama’s 2021 congressional redistricting. Alabama has seven congressional districts; the NAACP Legal Defense Fund and other plaintiffs argued that Alabama’s plan violated Section 2 by packing Black voters into a single majority-Black district (the 7th) while cracking the remaining Black population across six predominantly white districts, even though Black Alabamans constitute approximately 27% of the state’s population.

The district court (a three-judge court under 52 U.S.C. § 10304(a)) held that plaintiffs had demonstrated a likelihood of success on the merits and ordered Alabama to draw a new map with a second district in which Black voters had an opportunity to elect their preferred candidates. Alabama appealed; the Supreme Court issued a stay pending appeal.

The Supreme Court affirmed by a 5–4 vote.¹⁴ Chief Justice Roberts, writing for the majority, reaffirmed the *Gingles* framework as the operative standard for Section 2 redistricting claims. He rejected Alabama’s arguments that Section 2 required the drawing of majority-minority districts only to be evaluated through a race-neutral analysis, or that the Equal Protection Clause’s colorblindness requirement precluded a court from considering race in evaluating whether a second majority-minority district was required.

The majority’s core holding: *Gingles* means what it says. Where the three preconditions are met (geographic concentration and compactness, political cohesion, majority bloc voting), the totality of circumstances analysis must be conducted, and if it shows vote dilution, a remedial majority-minority district must be drawn. The Court reaffirmed that using

¹³*Allen v. Milligan*, 599 U.S. 1 (2023).

¹⁴Chief Justice Roberts joined the four-Justice liberal bloc in the majority; Justice Kavanaugh wrote a concurrence.

race as a factor in redistricting to comply with the VRA does not automatically violate the Equal Protection Clause—it is the use of race to *exceed* VRA requirements that triggers strict scrutiny.

Allen v. Milligan matters for the post-*Shelby* landscape in two ways. First, it confirms that Section 2 vote-dilution claims for redistricting remain viable and that courts will order remedial maps when plaintiffs prevail. Second, and more subtly, it demonstrates the power of the *Gingles* framework’s compactness analysis: the district court found, and the Supreme Court affirmed, that Alabama’s Black population was sufficiently concentrated to constitute a majority in a second congressional district—a finding driven by geographic and demographic analysis of the kind that algorithmic redistricting tools like BISECT can perform systematically.

4.2 State VRA Analogs

Several states have enacted state-law analogs to the VRA that provide additional protection beyond what federal law requires post-*Shelby*.

California Voting Rights Act (CVRA): Enacted in 2001 (Cal. Elec. Code §§ 14025–14032), the CVRA prohibits at-large elections that dilute minority voting rights and applies a lower standard than the federal *Gingles* test: it does not require a showing of geographically compact majority-minority district feasibility. The CVRA applies to local governments and school districts; it has generated substantial litigation and has resulted in numerous jurisdictions converting from at-large to by-district election systems.

New York Voting Rights Act (NYVRA): Enacted in 2022 as part of the John R. Lewis Voting Rights Act of New York (S51001-A), the NYVRA provides the most comprehensive state-level VRA analog in the country. It covers all levels of government, employs a results standard similar to the federal Section 2, and includes enhanced remedies. Critically, the NYVRA restores a preclearance-like mechanism for jurisdictions with recent voting rights violations, requiring them to obtain approval from the New York Attorney General before implementing changes to voting laws. This represents the most direct state-level substitute for federal Section 5 preclearance enacted anywhere in the country.

Washington Voting Rights Act (WVRA): Enacted in 2018 (Wash. Rev. Code § 29A.92), the WVRA prohibits dilution of minority voting rights in local elections and provides a results-based standard that applies broadly to at-large districts.

These state analogs vary in their applicability to congressional redistricting (which is governed primarily by federal law) vs. state and local redistricting (where states have broader authority). The NYVRA is the only state VRA analog that includes a preclearance-like

mechanism.

4.3 The Enforcement Asymmetry in Practice

Table ?? summarizes the enforcement asymmetry between the pre-*Shelby* and post-*Shelby* environments.

Table 1: VRA Redistricting Enforcement: Pre-*Shelby* vs. Post-*Shelby*

Dimension	Pre- <i>Shelby</i> (through 2013)	Post- <i>Shelby</i> (2013–present)
Primary tool	Section 5 preclearance	Section 2 litigation
Timing	Ex ante (before enactment)	Ex post (after enactment)
Process	Administrative (DOJ) or D.C. court	Federal district court
Burden	State proves non-retrogression	Plaintiff proves dilution
Timeline	Weeks to months	Years
Covered states	9 fully + parts of others	None (Section 4(b) invalid)
Standard	Non-retrogression	<i>Gingles</i> + totality

The enforcement asymmetry creates a strategic environment in which former covered states can enact potentially discriminatory plans knowing that any legal challenge will take years to resolve. The 2012 Texas congressional plan, challenged immediately after enactment, was not replaced until after the 2014 and 2016 elections; Alabama’s 2021 plan was not remedied until after the 2022 election (following *Allen v. Milligan*). In a world with functioning Section 5 preclearance, neither plan would have been implemented without federal approval.

5 Algorithmic Redistricting as Transparency Substitute

5.1 Section 5’s Prophylactic Function

Section 5 preclearance served a function that Section 2 litigation cannot replicate: it was a prophylactic rule that deterred discriminatory redistricting before it was enacted, not a remedy applied after the harm had occurred. Even jurisdictions that believed their redistricting plans were VRA-compliant had to justify that belief to the DOJ or a federal court before implementation. This review process had three effects:

1. **Deterrence:** Knowing that plans would be scrutinized ex ante discouraged covered jurisdictions from drawing maps that would fail the non-retrogression standard.

2. **Documentation:** The preclearance submission process required covered jurisdictions to document their redistricting criteria, methods, and demographic analysis. This created a contemporaneous record that could be used in subsequent litigation.
3. **Speed:** Administrative preclearance (the DOJ route) was available within 60 days; even the slower judicial preclearance route produced decisions before the election cycle. Section 2 litigation, by contrast, typically spans multiple election cycles.

All three of these functions disappeared with *Shelby County*. No deterrence exists in former covered states because no pre-enactment review is required. No documentation requirement compels contemporaneous disclosure of redistricting criteria and methodology. No fast-track process exists to resolve disputes before an election cycle begins.

5.2 Algorithmic Redistricting as Transparency Substitute

BISECT cannot restore Section 5 preclearance—that requires an act of Congress. But it can provide a transparency substitute that addresses two of Section 5’s three functions: deterrence (partially) and documentation (fully).

Documentation: The BISECT pipeline produces a complete SHA audit chain for every redistricting plan. Every input (Census tract data, adjacency graph), every intermediate result (bisection sequence, seed selection), and every output (district assignment file) is cryptographically hashed and chained. This produces a plan that is fully reproducible by any party with access to the public Census data and the published BISECT software. In expert witness proceedings, this means that the BISECT plan can be verified independently by opposing experts, eliminating disputes about whether the algorithmic result was manipulated.

Deterrence (partial): A BISECT reference plan available as a published neutral baseline before state legislatures begin drawing maps creates a deterrent that is weaker than Section 5 (it is not legally required) but stronger than nothing. Legislators and mapmakers who know that a neutral algorithmic plan will be produced and introduced into evidence must draw further from the neutral baseline to achieve a desired partisan or demographic result—and the further they draw, the more visible the deviation from the neutral plan becomes.

5.3 Three Use Cases in Post-*Shelby* Litigation

The BISECT reference plan serves three distinct roles in post-*Shelby* Section 2 redistricting litigation.

Use Case 1: Plaintiff Reference Plan. In a Section 2 vote-dilution case, the plaintiff must show not only that the enacted plan dilutes minority voting strength but that an

alternative plan exists that would provide adequate minority representation without sacrificing other redistricting criteria. Under *Allen v. Milligan*, the plaintiff’s burden includes demonstrating that a majority-minority district is “reasonably configured”—that is, compact and otherwise consistent with legitimate redistricting principles. A BISECT plan with VRA mode enabled (`--weights-override vra-aligned`) provides exactly this: a reasonably configured alternative plan that creates the required majority-minority district while satisfying population balance, contiguity, and compactness requirements. Because the plan is algorithmic and reproducible, it is resistant to the opposing argument that it was “drawn to prove a point” or constructed after the fact to win litigation.

Use Case 2: Defendant’s Neutral Map Defense. States defending their redistricting plans against Section 2 claims can introduce a BISECT reference plan to show that the enacted map does not deviate materially from what a neutral process would produce. If the BISECT plan and the enacted plan produce similar BVAP distributions, the defendant can argue that any deviation from a “perfect” minority representation outcome is attributable to geography rather than mapmaker intent. This defense is most compelling when the state ran BISECT as part of its redistricting process and adopted the plan with minimal modification—demonstrating that the neutral algorithm, not political manipulation, determined the outcome.

Use Case 3: Court-Appointed Special Master Baseline. When courts order remedial redistricting plans—as the Alabama district court did in *Allen v. Milligan* and the North Carolina Supreme Court did in *Harper v. Hall*—they often appoint special masters to produce alternative plans. BISECT is well-suited to this role: it produces plans quickly, satisfies the traditional redistricting criteria that courts require of remedial maps (population equality, contiguity, compactness), and can be configured to satisfy VRA requirements through the VRA mode. The full audit chain provides the transparency that a court-ordered plan requires.

5.4 The Non-Retrogression Analogy

Section 5’s non-retrogression standard asked: does the new plan reduce minority electoral opportunity relative to the existing plan? The BISECT reference plan enables an analogous comparison in Section 2 litigation: does the enacted plan reduce minority electoral opportunity relative to what a neutral, non-partisan process would produce?

This “neutral baseline” comparison does not replicate Section 5’s legal effect—Section 2 does not adopt a non-retrogression standard, and *Gingles* does not require comparison to a neutral baseline. But it provides a framework for the expert witness to quantify the

magnitude of minority vote dilution. A claim that a map “dilutes minority voting strength” is more compelling, and more legally actionable, when it can be attached to a specific quantity: “the enacted map produces a BVAP of X% in the relevant district, while a neutral algorithmic plan produces Y%, a gap of Z percentage points attributable entirely to mapmaker choices rather than demographic geography.” This quantification—the gap between the neutral baseline and the enacted plan—is the algorithmic redistricting contribution to post-*Shelby* Section 2 litigation.

6 Conclusion

Shelby County v. Holder changed the operational environment for minority voting rights in redistricting more than any other Supreme Court decision since *Thornburg v. Gingles* established the three-part test in 1986. By invalidating Section 4(b)’s coverage formula and rendering Section 5 preclearance a dead letter, the Court removed the most powerful and most efficient VRA enforcement tool: the ex ante administrative review that compelled covered jurisdictions to justify their redistricting plans before implementation.

Thirteen years later, the legislative fix—a new coverage formula—has not been enacted. Former covered states conduct redistricting without any federal pre-enactment review. Section 2 vote-dilution litigation under the *Gingles* framework remains the primary federal remedy, reinforced by *Allen v. Milligan*’s 2023 reaffirmation that the framework applies with full force to redistricting claims. Several states have enacted VRA analogs that provide supplemental protection, with New York’s John R. Lewis Voting Rights Act representing the most comprehensive restoration of preclearance-like review at the state level.

Algorithmic redistricting with BISECT fills part of the gap that Section 5’s demise created. It cannot restore the ex ante legal requirement that covered states prove non-retrogression; that requires Congressional action. But it provides the documentation function—a transparent, reproducible, verifiable neutral plan that any party can independently replicate and verify—and a partial deterrence function through the availability of a published neutral baseline. In Section 2 litigation, the BISECT reference plan quantifies the gap between what geography would produce under a neutral process and what the enacted map actually produces, translating the legal concept of “vote dilution” into a specific, measurable deviation from a neutral baseline.

Three practical conclusions follow for redistricting practitioners working in the post-*Shelby* environment:

1. **Run bisect early.** In former covered states with significant minority populations, produce and publish a BISECT reference plan before the legislative redistricting process

begins. This establishes the neutral baseline and, by making the deviation from that baseline visible, creates a deterrent against extreme redistricting.

2. **Document the VRA mode analysis.** The BISECT VRA mode (`--weights-override vra-aligned`) implements the *Gingles* three-prong analysis algorithmically. Run the VRA mode analysis for all relevant states and document the results: which districts are majority-minority under a neutral plan, what the BVAP thresholds are, and how the enacted plan compares. This documentation is the algorithmic equivalent of the Section 5 preclearance submission.
3. **Use the audit chain as evidence.** The BISECT SHA audit chain provides cryptographic proof that the reference plan was produced by the neutral algorithm and has not been modified. In litigation, this resolves disputes about whether the algorithmic plan was manipulated ex post to support a desired outcome. Present the audit chain as part of the expert witness report alongside the demographic and geographic analysis.

Section 5 cannot be restored by algorithmic means. But the transparency, reproducibility, and verifiability that algorithmic redistricting provides can substitute for the documentation function that Section 5 required, and can give Section 2 plaintiffs a quantitative, evidence-based framework for measuring the cost of minority vote dilution in the post-*Shelby* world.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author's own.